PlayGrid: Sparking AI Literacy with an Intuitive and Accessible Learning Interface

Richard Quiloan[1], Sanaz Rahimi Moosavi[1], Roman Tankelevich[1], and Mike Karlin[2]
[1]Department of Computer Science, California State University Dominguez Hills
[2]Department of Liberal Studies, California State University Dominguez Hills
Email: {rquiloan1, srahimimoosavi, rtankelevich, mkarlin}@csudh.edu

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***Abstract*—Artificial Intelligence (AI) has become increasingly commonplace for the average consumer, but AI is not commonly understood. In a global survey of adults, 1/3 of users thought they were using AI-powered services, but in reality, more than 3/4 of those consumers were using AI-powered technology. This uninformed but unrealized use of AI hinders peoples' capacity to effectively utilize, cooperate with, and operate as critical consumers of AI. There is a growing movement to develop AI literacy in young children. This paper demonstrates a way to introduce and familiarize young children with the Q-Learning algorithm using the Adafruit NeoTrellis as an introductory step in AI education. It is a fun, educational, and handheld gaming device that has the potential for other AI-powered games to support and increase AI literacy in young children.**

***Index Terms*—Artificial Intelligence, AI, AI literacy, early childhood education**

I. Introduction

Many people are unaware that they interact with artificial intelligence (AI) daily. In 6000 adults surveyed globally, only 33% of consumers thought they were using AI-powered services, but in reality, 77% of those consumers were using technology that featured AI [1]. The most common ways consumers use AI in their day-to-day lives are through personalized marketing, voice assistants, navigation and traveling, facial recognition technologies, and smart home devices [2]. Other pertinent ways AI is prevalent in people's everyday lives include Amazon product suggestions, Google auto-complete, and OpenAI's GPT-4 technology which is now integrated into Microsoft Bing [3]-[4]. Though AI has become increasingly commonplace, AI is not commonly understood [3]. This uninformed use of AI hinders peoples' capacity to effectively utilize, cooperate with, and operate as critical consumers of AI [5]. Furthermore, the AI industry is experiencing growing pains.

At the beginning of 2023, AI arrived at a pivotal and consequential juncture. In the previous months, OpenAI, an AI research and deployment company, launched ChatGPT (Chat Generative Pre-trained Transformer), a large language model (LLM) with human-like conversational interaction. It had a noted ability to answer follow-up questions, admit mistakes, oppose inaccurate assertions, turn down inappropriate requests, and communicate well-expressed responses over a wide range of disciplines [6]. Four months later, as ChatGPT released its fourth iteration, GPT-4, various AI and technology community leaders called for a six-month pause on training AI systems as powerful as GPT-4. The experts signed an open letter arguing these types of AI models posed significant risks to humanity and that "advanced AI could represent a profound change in the history of life on Earth, and should be planned for and managed with commensurate care and resources" [7]. The technology leaders claimed that the development of these AI models was happening too fast and that more time and resources were necessary to understand and mitigate the risks.

In response, other prominent AI developers and ethicists, including Microsoft co-founder Bill Gates, disagreed that a pause would be helpful [8]. Though Gates acknowledged more research was needed to understand the more complicated aspects of AI, he did not see how a temporary pause on the development of AI would solve any challenges or see how a suspension on AI development could be properly enforced. Furthermore, other experts reasoned that a moratorium on AI research could hinder progress in a rapidly evolving industry, potentially giving a competitive edge to authoritarian countries working on their own AI systems [9]. Coincidentally, a movement has been simultaneously emerging to help everyday people recognize and understand AI's role in their lives. More specifically, there is a growing movement to develop AI literacy in young children [10]–[13]. AI literacy can be defined as possessing the fundamental skills required for individuals not just to function but thrive in the ever-increasing AI-driven world [12].

In this paper, we propose a handheld learning interface that introduces the concept of AI literacy to young children. It does not teach coding related to AI. Instead, the primary purpose is to help children understand the capabilities and limitations of AI in a low-risk and educational environment. The nature of this paper is highly exploratory. The key contribution of this paper is to propose a novel and creative solution to support young children in exploring machine learning (ML)-based technologies.

This paper introduces three different prototypes – PlayGrid Mat, Sense, and Go. The prototypes use the Q-learning algorithm in a pathfinding game. The Q-learning algorithm is a model-free Reinforcement Learning (RL) algorithm that can learn optimal actions in complex environments without requiring a model of the environment. By iteratively updating its action-value, or Q-value, estimates, the Q-learning agent can eventually converge to an optimal policy, identifying the best actions to take in each state to maximize cumulative

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rewards [14].

II. Related Work

A. AI Literacy

Literacy empowers individuals to express their selves and communicate using written language [10]. It can enable an individual from a marginalized or oppressed group to question and challenge societal norms and transform their own lives and the world around them [15]. There have been current, growing efforts to teach literacy, primarily geared towards non-technical individuals, in various computer science-related skills known more broadly as computer science (CS) literacy [10], [16]. [10] detailed more specific categories under CS literacy, including *digital literacy*, which addresses proficiency in using computational devices, *computational literacy*, the ability to use code to articulate, investigate, and convey concepts, and *data literacy*, the ability to interpret, manipulate, examine, and debate using data within the broader context of exploring the world. The authors defined *AI literacy* as a "set of competencies that enables individuals to critically evaluate AI technologies; communicate and collaborate effectively with AI; and use AI as a tool online, at home, and in the workplace" [10]. The researchers argued that to have informed interactions with AI, it is a critical first step for people to recognize when AI is in use. Furthermore, accurately understanding the current capabilities, advantages, and limitations of AI enables people to make better-informed decisions about how to properly navigate an AI-entrenched environment [10].

In [16], researchers discovered that guiding 7 to 12-year-old children to develop custom machine learning models and programs improved their understanding of smart device limitations. The researchers found that the children "shifted their attribution of agency from the devices to the people who program them" [16]. Studies like [16] promote AI literacy in a way that empowers individuals to fully comprehend that they may exercise agency by making decisions and taking actions. At the same time, AI still influences those individuals' perceived autonomy over their decisions and actions [17]. [17] argued that the amount of agency and autonomy in relation to AI technology depends on factors like the transparency and accessibility of the technology, as well as the individual's own AI literacy. AI literacy is an emerging topic in digital literacy education research, as is in the following studies.

[12] identified key components of AI literacy and proposed a framework to help individuals navigate the AI-driven world effectively and responsibly. [11] described a Massachusetts Institute of Technology (MIT) AI summer workshop designed specifically for middle school students. [10] proposed a framework of AI literacy competencies and discussed design considerations for AI education. [17] identified AI literacy as one of four new ethical principles employed that are unique to K-12 education. Finally, [18] presented an innovative pedagogical model for AI literacy in early childhood education (ECE).

B. Toys and Games

Research in toys and games that teach programming to children as young as 3 years old is still in the early stages [19]. [19] argued there is still a comparatively small but growing amount of programming toys and games aimed at children from the ages of 3-8. A common format involves a robot that can be programmed to navigate through a grid-like maze using specific sequences to move forward or backward or turn left or right [20]. These toys and games are primarily used for developing skills in problem-solving, critical thinking, analytical thinking, directional concepts, visual tracking, and correlating 2-D and 3-D [19], but the process of planning a specific sequence of directions may be difficult for younger children. Additionally, when toys and games related to programming require the child to move their body and learning happens simultaneously, thoughts are connected to a physical action that expresses new knowledge, and a more powerful learning process occurs [21]. These types of toys are often used on the floor or a table and are referred to as tangible programming kits or tangible robot kits [19].

C. Reinforcement Learning

[22] described RL as a way of controlling agents through rewards and punishments without specifying how the task needs to be achieved. [23] suggested that "states are the observations that an agent or player sees and actions are the behaviors that the agent takes in those states." Essentially, states represent observations, while the actions are the agent's responses. RL aims to learn how to achieve the highest possible reward in the long term. Additionally, data is given in the form of state-action pairs. Other ML models are typically trained on fixed datasets. The agent does not know or need to label the object as an apple, for example. The agent simply learns that eating the apple prolongs its life, so when encountering an apple, the appropriate action is to eat it. The grid environment for the PlayGrid described throughout this research is highly suitable for the iterative learning process of RL, allowing agents to effectively engage in trial and error to optimize their strategies.

[24] stated that in RL, an agent, which can be the AI, learns from the environment by interacting with it through trial and error and receiving positive or negative rewards as feedback for performing actions. RL is "a computational approach of learning from actions" [24]. One of the main benefits of RL is that, without human guidance, the algorithm interacts with a dynamic environment where it performs different actions to explore the best way to accomplish its task [23].

For example, in the video game Snake, the length of the snake increases when it "eats" an apple. The "head" of the snake continuously moves in a direction without stopping until it is steered up, down, left, or right to avoid hitting the walls or its own body. In this game, the apple is the reward. In any given state, an agent sends outputs in the form of actions to the environment, and the environment will respond with new states and rewards if they exist. Furthermore, the reward may be delayed until the future. More information on RL can be found at [23] and [24].

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RL has made significant contributions in the gaming domain. It has been successfully applied to various types of games, from classic board games to more complex video games. [25] conducted groundbreaking research demonstrating the effectiveness of deep Q-networks (DQN) in playing Atari games. The RL agent in the study achieved human-level performance across several games.

1) Q-learning: [14] introduced Q-learning as a method for iteratively estimating the optimal action-value function. [23] stated that the Q-function takes two inputs – the agent's current state and the action executed while in that state. It returns the expected total future sum of rewards that an agent can receive after that point. If a really good action in the state is taken, the Q-function returns a high expected total future reward, while a bad action results in a low or penalized future reward [23].

Q-learning is a model-free reinforcement learning algorithm that can learn optimal actions in complex environments without requiring a model of the environment. By iteratively updating its action-value, or Q-value, estimates, the Q-learning agent can eventually converge to an optimal policy, identifying the best actions to take in each state to maximize cumulative rewards [14].

[26] explained that in pathfinding tasks, the agent can use the Q-learning algorithm to learn the shortest path or most efficient path between two points in an environment while considering various factors such as obstacles. The researchers argue that the algorithm's ability to learn from trial-and-error experiences and adapt to changing environments makes it suitable for pathfinding tasks in dynamic and unknown environments. There have been many modifications to the Q-learning approach to improve performance [27]–[30]. The Q-learning algorithm is widely used in pathfinding mazes and games [31], [26]-[28].

D. NeoTrellis

This paper introduces a novel implementation of the Q-learning algorithm in the NeoTrellis, a grid controller kit that is typically used in music production to trigger sounds, loops, and effects. At the time of writing, we found no published articles, dissertations, or studies utilizing the NeoTrellis, let alone utilizing the NeoTrellis using the Q-Learning algorithms. There were zero results for "NeoTrellis" in the ACM Digital Library, zero results in the IEEE Xplore Digital Library, and only two results in Google Scholar.

The Q-learning algorithm is well-suited for PlayGrid's grid environment, learning and adapting in dynamic settings. Through a "trial and error" approach, the agent or player can explore different actions and learn from the rewards that are received. In PlayGrid, the user can select a starting grid, target grid, and optional obstacle grids, creating a unique environment. By experiencing the benefits and limitations of Q-learning in navigation through the maze they have created, players gain insights into the strengths and weaknesses of ML algorithms, ultimately fostering AI literacy.

III. The Proposed Platforms

This paper started with an idea that evolved through three different prototypes. Each prototype proposes the potential for a system to introduce ML to young children.

The first prototype, PlayGrid Mat, was initially envisioned as a pallet on the floor with individually lit-up grids that users could interact with by stepping, jumping, or dancing on. Due to cost constraints, time-sensitive restrictions, and limited programming skills, PlayGrid Sense was created. PlayGrid Sense is more cost-effective and physically smaller, so a young child can hold it. After demonstrating a proof-of-concept for PlayGrid Sense, the lead author recognized that the interface and user interaction were not satisfactory and the human-computer interaction (HCI) component was weak. After more research, the NeoTrellis by Adafruit was selected to improve the system's HCI, leading to the third prototype, PlayGrid Go. It was more user-friendly and accessible to a young child. Though each subsequent prototype was in response to either the lead author's or component's limitation, each individual prototype has the potential to be its own full-fledged system.

A. PlayGrid Mat

1) PlayGrid Mat Design Goal: The hardware for PlayGrid was initially intended to be a sturdy 4x4 feet platform on the floor that users could jump, stomp, and dance on. It would need to be custom-built, which meant prohibitively high costs and an extended product development timeline that was unrealistic. Another challenging aspect was that the software would need to be written from scratch. Furthermore, PlayGrid Mat was intended to teach pre-coding skills by helping users build up to one sequential thought process many current coding toys and games start off with.

The academic literature did not have papers similar to PlayGrid, so the author referenced guides and tutorials from Makers or the Maker Movement, also known as the DIY (Do-it-yourself) community or How-to community [32]-[34].

[35] explained that these communities of tinkerers, hackers, and crafters started from the hobbyist movement and are made up of a loose collection of computer enthusiasts, artists, designers, and developers. The researchers also stated that open hardware platforms such as the Raspberry Pi and Arduino, popular crowd-funding websites like Kickstarter, the global spread of hacker- and makerspaces, publications such as Make magazine, and DIY websites such as Instructables, Hackaday.io, and Adafruit have all contributed in some way to a solid infrastructure for these creative communities.

Leeming created Unruly Splats [36], a durable, pressure-sensitive, plastic tile with sounds that children can program to light up or turn off when touched or stepped on [37]. [37] explained that children essentially program with their body movements by running around and jumping on the tiles to play games. They can be programmed by users as young as the age of 6 using blocks-based programming on tablets or computers [38]. Unfortunately, Unruly Splats are only available as an annual membership. At the time of this writing, the lowest price available was for 24 Unruly Splats for $4500.00/year (C.

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Webber, personal communication, Sept. 27, 2022), making it inaccessible to many. One of Leeming's early prototypes was a custom-made 4x4 grid on wooden studs, which seemed like a solid example of what PlayGrid Mat could be.

There is an emerging market of what [37] called computational kits and toys. They are technologies designed to enable children to explore computational concepts and practices.

creed_bratton_, a Maker, created a pressure-sensitive LED floor [39] using food cutting boards, 2x4 lumber frames, force-sensitive resistors, NeoPixel rings, and the microcontroller board, Arduino Mega 2560 [40]. This was another excellent example of what PlayGrid Mat could be. Unfortunately, the total cost for components was close to $2000.00 [39], which was too costly for this unfunded research work.

Utilizing tutorials from [39], the author created an initial prototype using WS2812B RGB LED strips, including two 8x8 attached matrices. [41]-[44] demonstrated that these LEDs could be cut, spliced, and reconnected with soldering wire. Each specific LED could then be controlled. After sketching a prototype and testing the proof of concept on the WS2812B RGB LED strips, the author recognized the potential ballooning of costs and time. To work within a budget and time constraints, PlayGrid Mat was modified into a smaller, handheld device.

Two WS2812B RGB 5050SMD 8x8 3.1in x 3.1in LED Matrixes were purchased and soldered together to create an 8x16 128 Pixel LED Matrix. The author was able to successfully run test programs on the matrix, but a critical feature was missing. The WS2812B had no way for users to interact with the program. A 2.4 GHz Wireless USB Controller was connected to the RPi for user input, but the interface was not user-friendly, and HCI was poor. After much trial and error, PlayGrid Mat's development ceased, and prototype 2, PlayGrid Sense, was created.

2) PlayGrid Mat Components:

a) Raspberry Pi 3 Model B+: The Raspberry Pi is a low-cost computer that runs Linux. It provides a set of GPIO (general purpose input/output) pins that allow users to control electronic components for physical computing and explore the Internet of Things (IoT). The RPi was used in each prototype.

b) WS2812B RGB LED: The WS2812B is an individually addressable LED light source that has the control circuit and RGB chip directly integrated into a 5050 RGB LED. Each addressable LED has an LED driver integrated circuit (IC) that enables each LED's brightness and color to be controlled individually [45]. The IC allows communication via a one-wire interface, meaning many LEDs can be controlled using just one digital pin from a microcontroller or SBC. The WS2812B matrix is essentially the WS2812 RGB LED strip mentioned above but connected in a matrix pattern.

c) 2.4 GHz Wireless USB Controller: The 2.4 GHz Wireless USB Controller is compatible with RPi.

3) PlayGrid Mat Architecture: The PlayGrid does not have a screen. Users interact with the device by stepping on the grids or by a wired joystick controller or 2.4GHz wireless USB controller. Figure 1 shows PlayGrid Mat's System Data Flow Diagram.

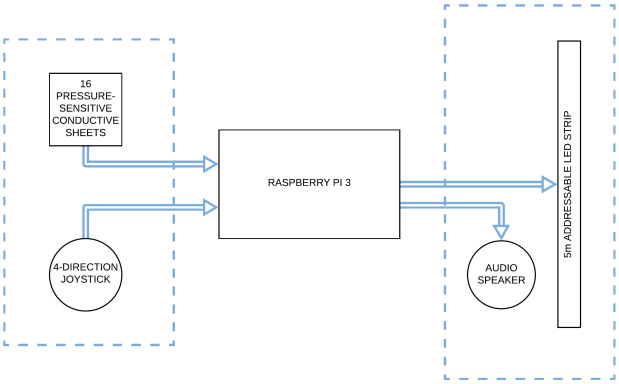

Fig. 1. PlayGrid Mat's System Data Flow Diagram

B. Prototype 2: PlayGrid Sense

1) PlayGrid Sense Design Goal: PlayGrid Sense was developed due to budget constraints, a growing development timeline, and programming limitations. This prototype would still include the same concepts of making ML accessible to young children but in a much smaller format that could be held in the hands. Instead of creating this prototype of PlayGrid from the ground up, an RPi Sense HAT was integrated into the system. Though the Sense HAT has many sensors, the main feature that was used for this paper was the 8x8 RGB LED matrix. The Sense HAT hardware was incredibly easy to install on the RPi. It mirrors the 40 GPIO pins and is simply secured directly on top of the RPi, as in Figure 2. Additionally, the RPi Sense HAT package contained existing libraries, making it quick and easy to begin using out of the box.

Kinsley's online tutorial [46] entailed creating a grid environment called *Blobworld* that utilized the Q-Learning algorithm. The environment had individually colored grids referred to as "blobs." The player blob is blue, the food blob is green, and the enemy blob is red. The main goal of this program is for the blue player blob to navigate its way as quickly as possible to the green food blob while simultaneously avoiding the red enemy blob. Kinsley recommends limiting the environment to between 10x10 and 20x20. Environments larger than 20 x 20 result in a larger Q-Table that takes up more memory and takes longer for the model to learn. As an example, a 10x10 Q -Table takes up about 15MB, and a 20x20 takes up about 195MB. In Kinsley's program, the blobs move in four diagonal directions – top-left, top-right, bottom-right, and bottom-left.

In this specific implementation, the agent is a "blob" that must navigate a 2D grid to find food while avoiding enemies. The Q-table is initialized with random values, and the agent takes actions based on the highest Q-value for a given state. The Q-values are updated based on the rewards received and the expected future rewards.

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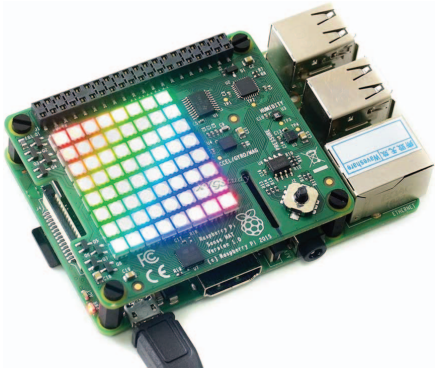

Fig. 2. PlayGrid Sense example

Fig. 3. PlayGrid Go

After successfully running the simulated environment on a monitor screen, Kinsley's code was modified to display simultaneously on the Sense HAT's 8x8 LED matrix. The goal of PlayGrid Sense was to create a proof of concept to show that the Q-Learning algorithm could be displayed on the Sense HAT, and this was accomplished. Unfortunately, after successfully creating the proof of concept, the HCI was still lacking. The Sense HAT's LED matrix only displayed what was in the simulated environment, but the user was not interacting with the hardware in any meaningful way.

During set up, the game Snake was programmed onto the RPi and Sense HAT, utilizing the Sense HAT's joystick. It inspired the author to search for other hardware that a user could interact with in a more purposeful manner.

2) PlayGrid Sense Components:

a) Raspberry Pi Sense HAT: The Raspberry Pi Sense HAT is an add-on board that gives the Raspberry Pi various sensing capabilities. The onboard sensors can monitor pressure, humidity, temperature, color, orientation, and movement [47]. It includes an 8x8 RGB LED matrix that helps present data from the sensors and a five-button joystick for interaction. It is compatible with any RPi computer with a 40-pin GPIO header. The Sense HAT was originally developed for use on the International Space Station [47].

C. PlayGrid 3: PlayGrid Go

1) PlayGrid Go Design Goal: The NeoTrellis by Adafruit is an excellent hardware choice for PlayGrid because it is essentially the handheld version of the initial vision of PlayGrid Mat. It has buttons that can be pressed for input and that have the ability to light up. The buttons are soft silicone, which feels welcoming and comfortable under the finger when pressed. The NeoTrellis is a grid controller kit and is typically used in music production to trigger sounds, loops, and effects.

The NeoTrellis kit contained everything, including the acrylic pieces for the case, RGB driver printed circuit board (PCB) for the 4x4 keypads, silicone elastomer 4x4 button keypads, a Feather M4 Express microcontroller board, and an I2C (Inter-Integrated Circuit) STEMMA cable [48].

After building the NeoTrellis kit, as seen in Figure 3, the simulation software was utilized from [49] "An autonomous path finding strategy for an Artificial Intelligence enabled Logo Mindstorm robot". The simulation software consists of the environment and the Q-Learning agent. [49] explained that the simulation software has a grid map with a matrix S of n * m where n is the number of rows and m is the number of columns. The max size can be 10x10, but for the purposes of this paper, the max defaulted to 8x8, the max LED matrix for the NeoTrellis, as seen in Figure 4.

The Feather M4 Express with CircuitPython was initially utilized. One of the primary benefits of using the Feather is that the program can run automatically once it is plugged in. This made the device more portable since all it requires is a 5V plug adapter. The lead author was able to run a drawing program on the NeoTrellis. Unfortunately, any Q-Learning programs ended in "limited memory" errors due to the small 512KB Flash memory on the Feather. Additionally, packages needed to be downloaded directly on the Feather, which also strained available resources. Resources for connecting the NeoTrellis to the RPi were scarce, but connecting four wires from the NeoTrellis I2C STEMMA socket to four GPIO pins – VIN to 5V, GND to GND, SDA to I2C data SDA, and SCL to I2C clock SCL – was a success.

After successfully running the simulation software on-screen, the code from [49] was modified to display simultaneously on the NeoTrellis. This process also involved multiple rounds of trial and error and learning and integrating many new libraries and packages. Finally, simulation software from [49] was displayed on the NeoTrellis.

When the program is run, the simulation software shows on the monitor and the user chooses the size of the grid (up to 8x8). The user then selects the *Start*, *Target*, and *Obstacle* grids. When the *Generate size* button is selected, the NeoTrellis lights up the appropriate amount of $n * m$ buttons. Both the simulation software and the NeoTrellis will show the Target grid exploring its respective matrix.

2) PlayGrid Go Components:

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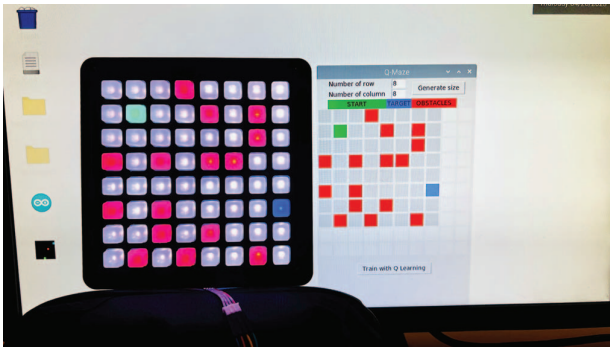

Fig. 4. PlayGrid Go running with simulation software

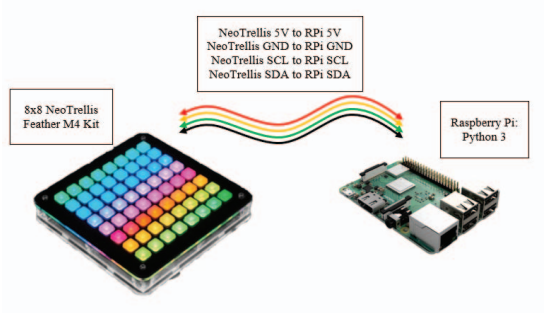

Fig. 5. PlayGrid Go architecture

a) Feather M4 Express: Feathers are a family of small, cross-compatible, and powerful microcontroller main-boards created by Adafruit [50], an online distributor of electronic parts and kits founded by Limor Fried [34].

b) NeoTrellis: The NeoTrellis was created by Adafruit, an online marketplace and educational hub for electronics for makers of all ages and skill levels [51]. The model used in this system is the 8x8 NeoTrellis Feather M4 Kit Pack. It is an open-source 8x8 Grid Controller Kit with four panels of 4x4 NeoTrellises enclosed by an acrylic laser-cut encasement. It communicates over I2C (alternatively known as I2C or Inter-Integrated Circuit) with the Feather M4 mainboard, although it is also possible for the NeoTrellis to communicate with the RPi and other SBCs. The four 4x4 button pad boards were soldered together by the lead author, and the four silicone elastomer 4x4 button keypads were placed on top. It is possible to make the NeoTrellis portable for untethered projects by installing a 3.7v 500 mAh lithium-ion polymer battery and mini panel mount DPDT toggle switch [48].

The NeoTrellis was inspired by Novation, created in 2009 as an 8x8 grid of large, illuminated buttons that could trigger sounds, loops, and effects. It revolutionized the electronic music hardware market [52]. Grid controllers are not a very popular tool outside of music production, but there are some individuals who have tried to create games using the NeoTrellis. [53] describes the process of using an 8x8 NeoTrellis to create a "Pixel Image Drawing Board" similar to the LED-Brite painting application in this paper.

c) PlayGrid Go's Architecture: PlayGrid Go's architecture in the paper is shown in Figure 5. It includes the required software and hardware components to build the prototype.

3) Conclusion: PlayGrid Mat, Sense, and Go are each their own system that introduce AI literacy to young children. Each prototype can serve as its own system or product. Existing Q-learning programs can be used and modified so players can interact with the RL algorithm.

IV. Results and Discussion

This research was a journey of exploration. Existing products were utilized in novel ways to place ML algorithms at the fingertips of young children. Each prototype offered insight into its own capabilities and limitations, yet the limitations directly inspired more creative ways that contributed to the solution.

Many existing CS games involve programming robots to navigate through grid-like mazes using specific sequences to move forward or backward or turn left or right [54]. These toys and games are effective in developing skills in problem-solving, critical thinking, and analytical thinking, but they may be more difficult to access for younger children around the age of three. This paper was directly inspired by these games that teach CS concepts and by non-academia-related projects created by individuals in the Maker community.

PlayGrid Mat's initial main goal was to help users build up to the sequential thought process many current coding toys and games start off with. For example, Learning Resources, an educational toy company, sells several screen-free, STEM (science, technology, engineering, and mathematics) coding toys for toddlers [55]. In the Code & Go Robot Mouse and Blue-Bot's simulation mode on an iPad, the user programs the sequences of steps, presses the *Go* button, then watches as the mouse or robot performs each sequence to find its way to the target. Trying to program the sequence seemed like it could be difficult for children around the age of 3. PlayGrid Mat would enable the user to step on one grid, then have the subsequent grid light up for the next grid in sequence.

Another creative product was developed by one of Seymour Papert's students, Marina Bers. She co-founded the DevTech research group at Tufts University, which created the visual programming language, ScratchJr, and the KIBO robot kit. ScratchJr is geared towards children ages 5 to 7 and is a spin-off of Scratch which teaches coding to children ages 8 to 16. KIBO is a fun, screenless, and user-friendly robot that young children can program by scanning sequential instructions to use in creative ways. Bers' research involves the design and study of innovative learning technologies to support children's positive development [56].

The Let's Go Code game has a similar concept, but instead of programming a robot, the child is the agent. The child must use a step-by-step approach to find an optimal pathway [57]. This is essentially a low-cost, analog version of PlayGrid Mat.

The attempt at PlayGrid Mat was a reminder that creating a product from scratch is a difficult, timely, and expensive pro-

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cess. It was challenging to progress without existing libraries to manage the movement of the LEDs to portray the grid maze environment. Addressable LED libraries do exist that display the typical light patterns of LED strips, but what currently exists were not appropriate for the purposes of this paper.

PlayGrid Sense's pre-existing libraries for the 8x8 LED matrix grid on the Sense HAT worked well. Kinsley's [46] tutorial was an excellent introduction to creating Blobworld, the Q-learning pathfinding environment. The prototype was created successfully by emulating what was on screen with the 8x8 LED matrix. The lack of user input led to the development of PlayGrid Go.

PlayGrid Go is essentially the handheld version of PlayGrid Mat. The simulation environment program from [49] worked well with the NeoTrellis. Though the development of the PlayGrid Go in this paper did not include user input from the NeoTrellis, with enough resources, time, and skill, this could be achieved.

All three PlayGrid prototypes offer real-world, practical, and novel ways to introduce AI literacy to young children. Though PlayGrid does not explicitly teach coding, programming, or AI or ML algorithms, it helps players understand the strengths and weaknesses of the Q-learning algorithm. As AI-related technology increases and becomes more integrated into everyday life in ways people see and do not see, it is important for individuals to recognize when and where AI exists so that they may be empowered to decrease any unwanted influences of AI but also enjoy the benefits AI can bring to communities.

V. Conclusion and Future Work

This paper demonstrates that an ML algorithm can potentially support AI literacy development for young children. The three separate prototypes can plant seeds of ideas of novel ways to expose young children to ML. This paper lays an initial framework for developers and educators to create a new tangible programming kit for young children and, really, people of all ages to interact with ML algorithms through fun, dynamic, and educational lo-fi games. It can be used as a guide to set up and develop more interactive and educational games. Additionally, this work opens opportunities to explore alternative ways to develop and interact with ML as well as explore other innovative pathways toward AI literacy.

The main focus of this paper was to establish a proof of concept – create an engaging, fun, and simple system that a young child could access and that utilizes Q-Learning to promote the initial development of AI literacy. The main factors that limited the paper included time constraints, high costs, and the author's limited experience programming in Python and working with unfamiliar hardware. Future research could utilize the Deep Q-Network (DQN) algorithms for speedier pathfinding. Additionally, the guide from [58] gives optional instructions for making the NeoTrellis fully wireless. There is enough space in the NeoTrellis for a 2500 mAh lithium-ion polymer (LiPoly) battery and an on-and-off switch. This would make PlayGrid Go handheld and wireless. Also, to improve the user interface, it would be outstanding if the user could operate the system without the use of any simulated environments. Lastly, there is a great deal of potential to create an "app store" of games on the NeoTrellis. Developers could create simple games utilizing Q-Learning or other ML algorithms to create a fun way for people to increase their AI literacy.

As people's lives become braided and intertwined with AI and ML, strategies, tools, and games like PlayGrid become increasingly important for AI literacy. By supporting AI literacy, people can be empowered to operate competently and strategically through AI-powered environments and settings. AI literacy helps people understand that even though their autonomy is influenced by AI, they can still make decisions that express their purest intentions.

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